FOR A SMOOTH AFFINE MORPHISM WITH GEOMETRICALLY IRREDUCIBLE FIBRES THE FUNCTOR OF FLATNESS IN DIMENSIONS AT LEAST D IS A FINITE DISJOINT UNION OF THE FUNCTORS OF FIBREWISE FREENESS OF A GIVEN RANK

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project own definitions of the reference, slogan and history display environments (verbatim package plus three newenvironment lines) that the retained file carries at its end and that the project itself typesets as comments. No macro was invented or redefined, and the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 05UE (sources/stacks-flat/pinned-original-source.tex line 7778).

Allowed standard prerequisites (editorial).

- The functor F_d of flatness in dimensions at least d and the functors H_p of fibrewise freeness. Same chapter, flat.tex, Equation labelled equation-flat-dimension-n (source line 6325) and Equation labelled equation-free-at-generic-points (source line 6166), both invoked in the statement. Fix the two functors whose comparison is the content of the lemma.
- Flatness at a point of the fibre gives freeness on an open neighbourhood of the generic point of the fibre. More on Morphisms, Lemma labelled more-morphisms-lemma-flat-and-free-at-point-fibre (Stacks tag 080Q), invoked at proof lines 7800-7802. Supplies the open neighbourhood U on which F_T is free of rank r, which is the only nontrivial input; the identification of the two functors is carried out in the counted span.

Source reference. Original equation 05MU: sources/stacks-flat/pinned-original-source.tex, source lines 6325–6325. Same-chapter pointer into the author environment equation at source line 6325; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original equation 05MR: sources/stacks-flat/pinned-original-source.tex, source lines 6166–6166. Same-chapter pointer into the author environment equation at source line 6166; invoked at proof line None of the counted

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proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 080Q: sources/stacks-chow/tags.txt, source lines 10190–10190. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 7802 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Assume that $X \to S$ is a smooth morphism of affine schemes with geometrically irreducible fibres of dimension d and that $\mathcal{F}$ is a quasi-coherent $\mathcal{O}_X$-module of finite presentation. Then $F_d = \coprod_{p=0,\ldots,c} H_p$ for some $c \geq 0$ with F_d as in (05MU) and H_p as in (05MR).*

Proof. As X is affine and $\mathcal{F}$ is quasi-coherent of finite presentation we know that $\mathcal{F}$ can be generated by $c \geq 0$ elements. Then $\dim_{\kappa(x)}(\mathcal{F}_x \otimes \kappa(x))$ in any point $x \in X$ never exceeds c. In particular $H_p = \emptyset$ for $p > c$. Moreover, note that there certainly is an inclusion $\coprod H_p \to F_d$. Having said this the content of the lemma is that, if a base change $\mathcal{F}_T$ is flat in dimensions $\geq d$ over T and if $t \in T$, then $\mathcal{F}_T$ is free of some rank r in an open neighbourhood $U \subset X_T$ of the unique generic point ξ of X_t. Namely, then H_r contains the image of U which is an open neighbourhood of t. The existence of U follows from More on Morphisms, Lemma 080Q. $\square$